MILITARY SPECIFICATION
YARN, ROVING, AND CLOTH: HIGH MODULUS, ORGANIC FIBER

1. SCOPE

1.1 Scope. This specification covers organic fibrous materials which will be used as yarns, rovings, and cloth used by themselves or as components of other materials. Treatments to provide compatibility with resin systems are included if applicable.

1.2 Classification. High modulus, continuous organic fibers shall be furnished in the forms and types as specified (see 6.1) below:

Form 1 - Yarn
Form 2 - Roving
Form 3 - Cloth
Type a - Untreated
Type b - Epoxy compatible
Type c - Polyimide compatible

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-1133 Packaging and Packing of Synthetic Fiber Fabrics

STANDARDS

FEDERAL

FED STD 191 Textile Test Method

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes

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MIL-STD-129 Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D2343-67 Tensile Properties of Glass Fiber Strands, Yarns and Roving Used in Reinforced Plastics

ASTM D1505 Density of Plastics by the Density-Gradient Technique

(Applications for copies of American Society for Testing and Materials specifications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Definitions.

3.1.1 Filament. A filament is defined as a rod of very small diameter drawn continuously to indefinite length.

3.1.2 Yarn. A yarn is defined as a multiplicity of filaments drawn together and gathered into an approximately parallel arrangement with or without twist.

3.1.3 Roving. A roving is defined as a number of yarns gathered together into an approximately parallel arrangement without twist.

3.1.4 Lubricant. A lubricant is defined as that minute quantity of organic or inorganic substance used to coat (surround) the fibers in order to keep the individual fibers apart and thus minimize abrasive action.

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3.1.5 Twist. The twist is defined in terms of number of turns per inch. Its direction is indicated by the letters "S" or "Z". Yarn structure has S-twist if, when held in a vertical position, the spirals conform in slope to the central portion of the letter "S", it has Z-twist if the spirals conform in slope to the central portion of the letter "Z".

3.1.6 Wrap and Filling (Ends and Picks). The construction of cloth or fabric consists of interweaving warp yarns (lengthwise of the fabric in the loom) and filling yarns (crosswise of the fabric in the loom). When defining a fabric construction, the wrap yarns are called ends and may be counted as ends per inch in wide goods, or as total ends when a width is stated for narrow goods. The filling yarns are called picks and are usually counted as picks per inch.

3.1.7 Denier. Denier is the weight in grams of 9000 meters of filament yarn or roving.

3.2 Design and Construction

3.2.1 Form 1 Yarn

3.2.1.1 Continuous-Filament Yarn. Continuous-filament yarn shall be furnished in conformance with Table I.

3.2.1.2 Tolerances. The following tolerances will be permitted on organic fiber yarn:

a. Denier plus or minus 10 percent.

b. No individual break shall be below 80 percent of the minimum requirements.

3.2.1.3 Twist. Yarn shall be furnished with less than 1 turn per 0.5 foot of twist unless otherwise specified by purchaser.

3.2.1.4 Put-Up. Yarn shall be wound on tubes or cones. Unless otherwise specified, the manufacturer's usual quantity of yarn per tube or cone will be acceptable.

3.2.1.5 Treatment. Material supplied shall be treated if required to insure compatibility with the resin system with which it is to be laminated.

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TABLE I

Yarn Form	1A	1B	1C	1D
Denier	195	380	1140	1520
Filaments per yarn, maximum	134	267	768	1000
Filament diameter, inches		.00046 ± .00005		
Density, gram/cm^3		1.45 ± 0.05		
Ultimate Tensile Strength, min., psi		490,000		
Modulus of Elasticity, min., 10^6 psi		17.5		

3.2.1.6 Type. Treatments shall be specified by the purchasing agency.

3.2.2 Form 2 Roving

3.2.2.1 Roving. Roving shall be furnished in conformance with Table II.

3.2.2.2 Tolerances. The following tolerances will be permitted on roving:

a. Denier and yield - plus or minus 10 percent.

b. No individual break shall be below 80 percent of the minimum requirements.

3.2.2.3 Twist. Roving shall be furnished without twist.

TABLE II

Roving Form	2A	2B
Roving Denier	4560	7100
Yield, grams/yard	0.463	0.721
Tensile Strength, min., psi	490,000	450,000
Modulus of Elasticity, min., 10^6 psi	17.5	17.5

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3.2.2.4 Put-Up. Roving shall be wound on tubes. Unless otherwise specified, the manufacturer's usual quantity of roving per tube will be acceptable.

3.2.2.5 Treatment. Material supplied shall be treated if required to insure compatibility with the resin system with which it is to be laminated.

3.2.2.6 Type. Treatments shall be specified by the purchasing agency.

3.2.3 Form 3 Cloth

3.2.3.1 Cloth Continuous-Filament. Cloth fabricated from continuous-filament yarn shall be furnished in conformance with table III.

3.2.3.2 Tolerances. The following tolerances will be permitted on cloth:

a. Count (ends and picks) -- plus or minus 2.

b. Ounces per square yard -- plus or minus 10 percent.

c. Width -- plus or minus 1/4 inch per yard of width.

d. Thickness --

Tolerances (inch)	Continuous Filament
0.003 inch and under	±0.0005
Over 0.003 to 0.011 inch	±0.001
Over 0.011 to 0.015 inch	±0.002
Over 0.015 inch	±0.003

3.2.3.3 Put-Up. Cloth shall be furnished in rolls, in quantities shown in Table III, and shall be spiral or convolute wound on tubes measuring 3 inches inside diameter and at least 3/4 inches longer than the overall width of the cloth. No roll shall contain more than three (3) pieces and no piece shall be less than 15 yards in length.

3.3 Identification Tickets. Each tube, cone, or roll of organic fiber material shall have an identification gummed label attached. The color of the label shall be light in intensity to permit easy

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TABLE III

CONSTRUCTION OF WOVEN ORGANIC CLOTH

Cloth Style	Ends and Picks		Yarn Type* Warp	Fill	Weave	Basis g/m^2	Weight (oz/yd^2)	Thickness mm	(in)	Width cm	(in)
120	34	34	195-1/0	195-1/0	Plain	61.1	1.8	0.114	0.0045	96.5	38
220	22	22	380-1/0	380-1/0	Plain	74.7	2.2	0.127	0.005	96.5	38
181	50	50	380-1/0	380-1/0	8-H Satin	169.8	5.0	0.279	0.011	96.5	38
281	17	17	1140-1/0	1140-1/0	Plain	169.8	5.0	0.254	0.010	96.5	38
13	16	46	380-1/0	380-2/0	Plain	176.6	5.2	0.432	0.017	45.7	18
143	100	10	380-1/0	380-1/0	4-H Satin	190.2	5.6	0.330	0.013	96.5	38
328	17	17	1420-1/0	1420-1/0	Plain	220.7	6.5	0.356	0.014	96.5	38

*Denier - number of plies/twist (turns per inch)
**Between selvages

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reading of printed, stamped, or typed markings. The label shall be legibly printed with water insoluble ink and will include the following information:

Manufacturer's name
Manufacturer's identification
Nomenclature
Denier or yardage

Each container shall be legibly marked by label and stencil with water insoluble ink which will include the following information:

Manufacturer's name
Manufacturer's identification
Nomenclature
Specification number
Contract number and date
Name of contracting agency

3.4 Workmanship. The finished material shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Material. The fibers shall be consistent in cross-section, reproducible in chemical composition and morphology. The material shall be free from foreign particles, dirt, lint and other impurities.

4.2 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.3 Inspection. Inspection shall be in accordance with the provisions of MIL-STD-105, except where otherwise specified.

4.3.1 Examination of Yarn and Roving. The defects listed in Table IV shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition, in which

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case only the more serious defect shall be counted. The sample unit shall be 10 yards or 10 percent of the yardage of the cone or tube, whichever is the smallest quantity. The lot size for this examination shall be expressed in units of one cone or tube as applicable. The acceptable quality level shall be 0.65 major and 4.0 total defects (major and minor combined) defects per hundred units. The inspection level shall be level I.

4.3.1.1 Examination of Yarn and Roving for Total Weight Per Tube. The sample unit for this examination shall be one tube. The inspection level shall be level S-3 and the acceptable quality level shall be 4.0 defects per hundred units, except for lots consisting of 500 of fewer units, then sample size shall be 10 tubes and the lot shall be unacceptable if two or more tubes in the sample are defective in regard to weight. Any tube found not within the range of weight specified in 3.2.1.2 or 3.2.2.2 shall be considered a defect with respect to weight.

TABLE IV

VISUAL EXAMINATION OF YARN

		Classification	
		Major	Minor
Appearance & workmanship..	Cut, any.....	X	
	Chafed or damaged, affecting serviceability.....	X	
	Finish other than specified	X	
	Spot or stain*.....		X
	Embedded foreign matter*..		X
Put-up.....	Any defect affecting the free unhampered unwinding of yarn or affecting the secure holding of yarn or winds on the package.....	X	
	Not put-up on cone or tube as specified.....		X

4.3.2 Examination of Cloth. The defects found during the examination of the cloth, shall be scored in accordance with 4.3.2.1 through 4.3.2.3.

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*Clearly visible at normal inspection distance (approximately 3 feet).

4.3.2.1 Yard-by-Yard Examination of Cloth. The required yardage of each roll of cloth shall be examined on both sides for defects listed in Table V. All defects found shall be counted regardless of their proximity to one another, except where two or more defects represent a single local condition of the end item, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each lengthwise yard or fraction thereof in which it appears. The sample unit shall be one linear yard full width. The acceptable quality levels shall be 2.5 major and 10 total (major and minor combined) defects per hundred units (yards) for the cloth. The number of tubes or rolls, as applicable, from the sample is to be selected shall be in accordance with Table VI. An approximately equal number of yards shall be examined from each tube or roll selected.

4.3.2.2 Overall Examination of Cloth. Each defect listed below shall be counted no more than once in each tube or roll examined. The sample unit for this examination shall be one tube or roll, as applicable. The sample size and acceptance number shall be as shown in Table VI. The lot size shall be expressed in units of one (1) linear yard each.

a. Objectionable odor.

b. Overall uncleanliness.

c. Width other than specified.

d. Uneven weaving.

TABLE V

Lot Size in Yards	Sample Size in Tubes or Rolls	Maximum Number of Defect Acceptable
Up to 1300[1].....	3	0
1301 to 3200.....	5	0
3201 to 8000.....	7	0
8001 to 22,000.....	10	0
22,001 to 110,000.....	15	1
110,001 and over.....	25	1

[1]If a lot contains fewer than three (3) tubes or rolls, each tube or roll in the lot shall be examined.

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TABLE VI

VISUAL EXAMINATION OF CLOTH

Defect	Classification Major	Minor
Hole, cut, tear, or smash - any	X	
Baggy, ridgy, or wavy (cloth)	X	
Spot, stain, or streak*:		
2 inches or more in combined length and width directions	X	
Less than 2 inches in combined length and with directions		X
Tender or weak spot*	X	
Broken or missing yarn:		
3 or more continguous, regardless of length	X	
2 continguous, 2 inches or more in length	X	
2 continguous, less than 2 inches in length		X
Single, 2 inches or longer or full width if filling-wise direction is less than two inches		X
Float and Skip:		
Multiple, 2 inches or more in length	X	
Multiple less than 2 inches in length		X
Single, 2 inches or longer of full width if filling-wise direction is less than two inches	X	
Thick or thin place*		X
Crease	X	
Brittle or fused area - any	X	

*Clearly visible at normal inspection distance (3 feet).

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4.3.3 Examination of Preparation for Delivery Requirements. An examination shall be made to determine that packaging, packing, and marking requirements of Section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and AWL shall be 4.0 defects per 100 units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, inadequate stapling, improper taping. Bulging or distortion of containers.
Weight	Weight of contents exceeds requirements.

4.4 Testing of the end item. The methods of testing specified in Federal Test Method Standard 191, wherever applicable and as listed in Table VII, shall be followed. Except where otherwise indicated, the physical values specified in Section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample size shall be as follows:

Lot Size	Sample Size
800 or less	2
801 to 22,000	3
22,001 and over	5

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The lot shall be unacceptable if one or more units fail to meet any requirements specified. The lot size shall be expressed in units of 1 linear yard for cloth and 1 pound for yarn.

TABLE VII

TEST METHODS

	Requirement Paragraph	Test Method
Yarn:		
Turns twist	3.2.1.3	4054
Denier	3.2.1.2, Table I	4010
Tensile Strength	Table I	ASTM D2343-67
Tensile Modulus	Table I	ASTM D2343-67
Density	Table I	ASTM D1505
Roving:		
Turns twist	3.2.2.3	4054
Yield	3.2.2.2, Table II	4010
Tensile Strength	Table II	ASTM D2343-67
Tensile Modulus	Table II	ASTM D2343-67
Cloth:		
Weave	Table III	Visual(1)
Yarns per inch (minimum)		
Warp	Table III	5050
Filling	Table III	5050
Thickness	Table III	5030
Weight oz/yd^2	Table III	5041

(1) One determination per sample unit and results reported as "pass or fail".

5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Level A

Form 1 and 2. Each tube or cone shall be individually wrapped in a 30 pound minimum basic weight kraft paper or enclosed in a wrap or bag of clear polyethylene fiber, 0.0015 inch thickness minimum. The wrap or bag shall be secured with a piece of pressure sensitive tape.

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Form 3. Cloth, put up as specified, shall be packaged in accordance with the requirements of PPP-P-1132.

5.1.2 Level C. Fiber material put up as specified shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing

5.2.1 Level B

Form 1 and 2. Yarn or roving of one type only, packaged as specified shall be packed in a snug fitting shipping container conforming to style RSC, Type CF or SF, class domestic, variety SW, grade 275 of PPP-B-636. The weight of contents shall not exceed 65 pounds.

Form 3. Each fiber roll shall have a piece ticket as required by PPP-P-1133.

6. NOTES

6.1 Intended Use. Materials described in this specification are intended for use in fiber reinforced plastics, tension cables, and other specialty applications.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Form required (see 1.2).
Form 1: Denier required (see Table I)
Form 2: Denier or yield required (see Table II)
From 3: Style and construction required (see Table III).

c. Type required (see 1.2).
Type a: Untreated
Type b: Epoxy compatible
Type c: Polyimide compatible

Custodian:
Air Force - 11

Preparing Activity:
Air Force - 11

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